CMPG 322 - Decision Support Systems (DSS) 2

SU8 - Markov analysis: Introduction and State Probabilities



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Outline

- States and state probabilities
- Matrix of transition probabilities
- Predicting future market shares
- Markov analysis of machine operations
- Equilibrium conditions
- Absorbing states and the fundamental matrix

- Markov analysis deals with the probabilities of future occurrences by analyzing presently known probabilities.
- Assumes the system starts in an initial state or condition.
- The probabilities of changing from one state to another are called a matrix of transition probabilities.
- Requires basic matrix manipulation.

- Discussion limited to Markov problems that follow four assumptions:
  1. There are a limited or finite number of possible states.
  2. The probability of changing states remains the same over time.
  3. We can predict any future state from the previous state and the matrix of transition probabilities.
  4. The size and makeup of the system do not change during the analysis.

- States are used to identify all possible conditions of a process or system.
- Markov analysis assumes states are both collectively exhaustive and mutually exclusive.
- The next step is to determine the probability that the system is in this state.

- **Information is placed into a *vector of state probabilities***

$\pi(l)$ = vector of state probabilities for period $l = (\pi_1, \pi_2, \pi_3, ..., \pi_n)$

where

$$n = \text{number of states}$$

$\pi_1, \pi_2, ..., \pi_n$ = probability of being in state 1, state 2, ..., state n

- In cases with one item, it is possible to know with complete certainty in which state an item is in

Vector of states can then be represented as:

$$\pi(1) = (1, 0)$$

where

$$\pi(1) = \text{vector of states for the machine in period 1}$$

$$\pi_1 = 1 = \text{probability of being in the first state}$$

$$\pi_2 = 0 = \text{probability of being in the second state}$$

In most cases, we are dealing with more than one item.

Example: Grocery Store

- States for people in a small town with three grocery stores
- A total of 100,000 people shop at the three groceries during any given month

- **Forty thousand** may be shopping at American Food Store – **state 1**
- **Thirty thousand** may be shopping at Food Mart – **state 2**
- **Thirty thousand** may be shopping at Atlas Foods – **state 3**

Probabilities are as follows:

State 1 – American Food Store:	$40{,}000 \div 100{,}000 = 0.40 = 40\%$
State 2 – Food Mart:	$30{,}000 \div 100{,}000 = 0.30 = 30\%$
State 3 – Atlas Foods:	$30{,}000 \div 100{,}000 = 0.30 = 30\%$

These probabilities can be placed in the following vector of state probabilities:

$$\pi(1) = (0.4, 0.3, 0.3)$$

where

$\pi(1)$ = vector of state probabilities for the three grocery stores for period 1

$\pi_1 = 0.4$ = probability that person will shop at American Food, state 1

$\pi_2 = 0.3$ = probability that person will shop at Food Mart, state 2

$\pi_3 = 0.3$ = probability that person will shop at Atlas Foods, state 3

- The vector of state probabilities represents **market shares** for the three grocery stores
- Management will be interested in how their market share changes over time

Key Insight

The state probabilities $\pi(1) = (0.4, 0.3, 0.3)$ correspond to:

- 40% market share for American Food Store
- 30% market share for Food Mart
- 30% market share for Atlas Foods

Example: Grocery Store

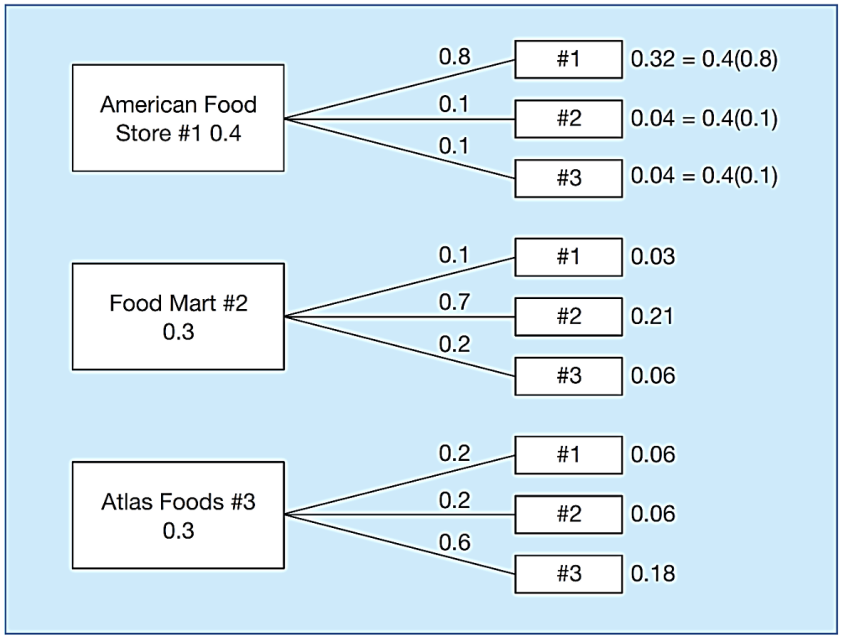


Figure: Tree Diagram for Grocery Stores Example

- Visual representation of state transitions over time
- Shows possible paths and probabilities between states
- Useful for understanding multi-period transitions

- A tree diagram for several months would get very large
- Easier to use a **matrix of transition probabilities**
- A **transition probability** is the probability of moving from one particular state to another particular state

Advantage of Matrix Approach

- More compact representation for multiple time periods
- Enables mathematical operations and predictions
- Scalable to larger systems with more states

- The matrix of transition probabilities allows us to get from a current state to a future state

Let P_{ij} = conditional probability of being in state j in the future given the current state of i

Example Interpretation

For example, P_{12} is the probability of being in state 2 in the future given the event was in state 1 in the period before

- P_{ij} represents the probability of transitioning **from** state i **to** state j

Matrix of Transition Probabilities

Let $P =$ the matrix of transition probabilities

$$P = \begin{pmatrix} P_{11} & P_{12} & P_{13} & \cdots & P_{1n} \\ P_{21} & P_{22} & P_{23} & \cdots & P_{2n} \\ \vdots & & & & \vdots \\ P_{m1} & P_{m2} & \cdots & & P_{mn} \end{pmatrix}$$

- Individual P_{ij} values are determined empirically
- The probabilities in each row will sum to 1
- m and n represent the number of states (often $m = n$)

Important Property

$$\sum_{j=1}^{n} P_{ij} = 1 \quad \text{for all } i$$

Each row sums to 1, representing all possible transitions from a given state.

Matrix of Transition Probabilities

- Use historical data to develop the matrix

$$P = \begin{pmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{pmatrix}$$

- Rows represent current state (from)
- Columns represent next state (to)
- Each row sums to 1: $0.8 + 0.1 + 0.1 = 1$, etc.
- Based on historical customer shopping patterns

Matrix of Transition Probabilities

$$P = \begin{pmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{pmatrix}$$

Row 1

- $P_{11} = 0.8$: State $1 \rightarrow$ State 1
- $P_{12} = 0.1$: State $1 \rightarrow$ State 2
- $P_{13} = 0.1$: State $1 \rightarrow$ State 3

Row 2

- $P_{21} = 0.1$: State $2 \rightarrow$ State 1
- $P_{22} = 0.7$: State $2 \rightarrow$ State 2
- $P_{23} = 0.2$: State $2 \rightarrow$ State 3

Row 3

- $P_{31} = 0.2$: State $3 \rightarrow$ State 1
- $P_{32} = 0.2$: State $3 \rightarrow$ State 2
- $P_{33} = 0.6$: State $3 \rightarrow$ State 3

Predicting Future Market Shares

- One of the purposes of Markov analysis is to predict the future
- Use the transition probability matrix to determine the state probabilities for a future date
- Allows the computation of the probability that a person will be at one of the grocery stores in the future
- Since this probability is equal to market share, it is possible to determine the future market shares of the grocery stores

Key Application

- Forecast customer behavior over time
- Predict changes in market competition
- Support strategic business decisions

- When the current period is 0, the state probabilities for the next period 1 are determined as follows:

$$\pi(1) = \pi(0)P$$

- For any period n, we can compute the state probabilities for period $n + 1$

$$\pi(n + 1) = \pi(n)P$$

Recursive Calculation

The state probabilities evolve over time by repeatedly multiplying the current state vector by the transition matrix.

The computations for the next period's market share are:

$$\pi(1) = \pi(0)P = (0.4, 0.3, 0.3)\begin{pmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{pmatrix}$$

$$= [(0.4)(0.8)+(0.3)(0.1)+(0.3)(0.2),\ (0.4)(0.1)+(0.3)(0.7)+(0.3)(0.2),\ (0.4)(0.1)+(0.3)(0.2)+(0.3)(0.6)]$$

$$= (0.32 + 0.03 + 0.06,\ 0.04 + 0.21 + 0.06,\ 0.04 + 0.06 + 0.18) = (0.41, 0.31, 0.28)$$

Matrix Multiplication

Refresh your matrix multiplication skills by viewing the video "Matrix multiplication"

- The market share for American Food and Food Mart have increased and the market share for Atlas Foods has decreased
- We can determine whether this will continue by looking at the state probabilities in the future
- For two time periods from now:

$$\pi(2) = \pi(1)P$$

Observed Changes

- American Food: 0.40 $\rightarrow$ 0.41 (+0.01)
- Food Mart: 0.30 $\rightarrow$ 0.31 (+0.01)
- Atlas Foods: 0.30 $\rightarrow$ 0.28 (-0.02)

Since we know that:

$$\pi(1) = \pi(0)P$$

We have:

$$\pi(2) = \pi(1)P = [\pi(0)P]P = \pi(0)PP = \pi(0)P^2$$

In general:

$$\pi(n) = \pi(0)P^n$$

Long-Term Behavior

The question of whether American and Food Mart will continue to gain market share and Atlas will continue to lose is best addressed in terms of equilibrium or steady-state conditions.

- The owner of Tolsky Works has recorded the operation of his milling machine for several years
- Over the past 2 years:
  - 80% of the time the milling machine functioned correctly for the current month if it had functioned correctly during the preceding month
  - 90% of the time the machine remained incorrectly adjusted if it had been incorrectly adjusted in the preceding month
  - 10% of the time the machine operated correctly in a given month when it had been operating incorrectly the previous month

Two-State System

- State 1: Machine functioning correctly
- State 2: Machine functioning incorrectly

Example: Tolsky Works

The transition probability matrix for this machine:

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.1 & 0.9 \end{pmatrix}$$

where

- $P_{11} = 0.8 =$ probability that the machine will be correctly functioning this month given it was correctly functioning last month
- $P_{12} = 0.2 =$ probability that the machine will not be correctly functioning this month given it was correctly functioning last month
- $P_{21} = 0.1 =$ probability that the machine will be correctly functioning this month given it was not correctly functioning last month
- $P_{22} = 0.9 =$ probability that the machine will not be correctly functioning this month given it was not correctly functioning last month

Example: Tolsky Works

- What is the probability that the machine will be functioning correctly 1 month from now?

$$\pi(1) = \pi(0)P$$

Assuming the machine is currently functioning correctly ($\pi(0) = (1, 0)$):

$$\pi(1) = (1, 0) \begin{pmatrix} 0.8 & 0.2 \\ 0.1 & 0.9 \end{pmatrix}$$

$$= [(1)(0.8) + (0)(0.1),\ (1)(0.2) + (0)(0.9)]$$

$$= (0.8, 0.2)$$

Result

Probability of functioning correctly in 1 month: 80%
Probability of functioning incorrectly in 1 month: 20%

Example: Tolsky Works

- What is the probability that the machine will be functioning correctly 2 months from now?

$$\pi(2) = \pi(1)P$$

Using $\pi(1) = (0.8, 0.2)$ from previous calculation:

$$\pi(2) = (0.8, 0.2)\begin{pmatrix} 0.8 & 0.2 \\ 0.1 & 0.9 \end{pmatrix}$$

$$= [(0.8)(0.8) + (0.2)(0.1),\ (0.8)(0.2) + (0.2)(0.9)]$$

$$= (0.64 + 0.02,\ 0.16 + 0.18) = (0.66, 0.34)$$

Result

Probability of functioning correctly in 2 months: 66%
Probability of functioning incorrectly in 2 months: 34%

Equilibrium Conditions

- It is easy to imagine that all market shares will eventually be 0 or 1
- Equilibrium share of the market values or probabilities generally exist
- Called **steady-state** or **equilibrium probabilities**
- An **equilibrium condition** exists if state probabilities do not change after a large number of periods
- At equilibrium, state probabilities for the next period equal the state probabilities for current period
- Equilibrium state probabilities can be computed by repeating Markov analysis for a large number of periods

Key Insight

Systems often reach a stable distribution where probabilities remain constant over time, regardless of the starting state.

State Probabilities

Period	State 1	State 2
1	1.000000	0.000000
2	0.800000	0.200000
3	0.660000	0.340000
4	0.562000	0.438000
5	0.493400	0.506600
6	0.445380	0.554620
7	0.411766	0.588234
8	0.388236	0.611763
9	0.371765	0.628234
10	0.360235	0.639754
11	0.352165	0.647834
12	0.346515	0.653484
13	0.342560	0.657439
14	0.339792	0.660207
15	0.337854	0.662145

Table: Probabilities are converging toward equilibrium values around (0.333, 0.667)

Equilibrium Conditions

It is always true that:

$$\pi(\text{next period}) = \pi(\text{this period})P$$

or

$$\pi(n+1) = \pi(n)P$$

At equilibrium:

$$\pi(n+1) = \pi(n)$$

So at equilibrium:

$$\pi(n+1) = \pi(n)P = \pi(n)$$

or

$$\pi = \pi P$$

Equilibrium Equation

The fundamental equation for finding equilibrium probabilities: $\pi = \pi P$

For Tolsky's machine:

$$\pi = \pi P$$

$$(\pi_1, \pi_2) = (\pi_1, \pi_2) \begin{pmatrix} 0.8 & 0.2 \\ 0.1 & 0.9 \end{pmatrix}$$

Performing the matrix multiplication:

$$(\pi_1, \pi_2) = [(\pi_1)(0.8) + (\pi_2)(0.1),\ (\pi_1)(0.2) + (\pi_2)(0.9)]$$

System of Equations

This gives us the following system:

$$\pi_1 = 0.8\pi_1 + 0.1\pi_2$$
$$\pi_2 = 0.2\pi_1 + 0.9\pi_2$$

The first and second terms on the left side, π_1 and π_2, are equal to the first terms on the right side:

$$\pi_1 = 0.8\pi_1 + 0.1\pi_2$$

$$\pi_2 = 0.2\pi_1 + 0.9\pi_2$$

The state probabilities sum to 1:

$$\pi_1 + \pi_2 + \cdots + \pi_n = 1$$

For Tolsky's machine:

$$\pi_1 + \pi_2 = 1$$

Complete System

We now have a complete system of equations to solve for the equilibrium probabilities.

We arbitrarily decide to solve the following two equations:

$$\pi_2 = 0.2\pi_1 + 0.9\pi_2$$

$$\pi_1 + \pi_2 = 1$$

Through rearrangement and substitution:

$$\pi_2 - 0.9\pi_2 = 0.2\pi_1$$

$$0.1\pi_2 = 0.2\pi_1$$

$$\pi_2 = 2\pi_1$$

Substitute into the sum equation:

$$\pi_1 + 2\pi_1 = 1$$

$$3\pi_1 = 1$$

$$\pi_1 = \frac{1}{3} = 0.33333333$$

$$\pi_2 = \frac{2}{3} = 0.66666667$$

- Examples so far assume it is possible to go from one state to another
- If you must remain in a state it is called an **absorbing state**

Absorbing State Definition

- A state from which it is impossible to leave
- Once entered, the system remains in that state forever
- Transition probability $P_{ii} = 1$ for absorbing states

Examples

- Bankruptcy states in financial models
- Terminal states in game theory
- Product retirement in market analysis

Questions

